

Discrete Choice Experiment Questionnaire: Vehicle Purchase Preferences

1. Screening Questions

Q1. Do you plan to purchase a vehicle within the next three years?

- ☐ Yes
- ☐ No
- ☐ Not sure

Q2. Even if you do not currently plan to purchase a vehicle, do you pay attention to automobile products or related information?

- ☐ Yes
- ☐ No

2. Instructions

In this survey, you will be presented with a series of hypothetical vehicle choice tasks.

In each task, you will be asked to choose between two vehicles (Vehicle A and Vehicle B).

Each vehicle is described by several attributes. Please read each option carefully and select the vehicle you prefer in each task.

The attributes are described as follows:

1. Purchase Price: The price paid by the buyer to purchase the vehicle.

2. Powertrain: The source of power for the vehicle.

- Battery Electric Vehicle (BEV): A vehicle powered by a battery system.
- Internal Combustion Engine Vehicle (ICEV): A vehicle powered by a conventional fuel engine.

3. Monthly Operating Cost: Typical monthly energy/fuel and routine running cost under normal use.

4. Daily Energy Replenishment Convenience : How easy it is to keep the vehicle ready for daily use through charging or refueling.

- Easy : Can be charged or refueled conveniently near home, work, or along usual routes.
- Moderate : Requires some planning or a short detour.
- Difficult : Requires advance planning, waiting, or a clear detour from usual routes.

5. Smart Cockpit Features : The level of digital interface and in-car software functionality.

- Basic : Small or standard screen, Bluetooth/audio functions, no advanced voice assistant.
- Intermediate : Larger touchscreen, stable smartphone integration, basic voice commands.
- Advanced : Large touchscreen, advanced voice assistant, personalized settings, and richer software functions.

6. Cabin Comfort : The quality of the in-cabin comfort experience.

- Basic : Standard seats and interior comfort.
- Improved : More comfortable seats and better cabin environment.
- Premium : High seating comfort, quiet cabin, and refined interior materials.

7. Long-term Reliability : Expected reliability and stability over several years of ownership.

- High : Few expected problems and strong long-term dependability.
- Medium : Occasional issues may occur but overall acceptable.
- Low : Higher likelihood of problems and weak long-term reliability.

3. Choice Tasks

Block A

TA1 (T01)

Attribute	Vehicle A	Vehicle B
Purchase price	RMB 110,000–130,000	RMB 110,000–130,000
Powertrain	Battery electric vehicle (BEV)	Gasoline vehicle (ICEV)
Monthly operating cost	About RMB 600 per month	About RMB 300 per month
Daily energy replenishment convenience	Moderate	Very convenient
Smart cockpit features	Advanced	Intermediate
Long-term reliability	Medium	Medium
Cabin comfort and entertainment experience	Improved	Improved

Which vehicle would you choose?

- A
- B

TA2 (T02)

Attribute	Vehicle A	Vehicle B
Purchase price	RMB 150,000–170,000	RMB 150,000–170,000
Powertrain	Battery electric vehicle (BEV)	Gasoline vehicle (ICEV)
Monthly operating cost	About RMB 600 per month	About RMB 300 per month
Daily energy replenishment convenience	Difficult	Easy
Smart cockpit features	Intermediate	Advanced
Long-term reliability	Medium	Low
Cabin comfort and entertainment experience	Improved	Improved

Which vehicle would you choose?

- ☐ A
- ☐ B

TA3 (T03)

Attribute	Vehicle A	Vehicle B
Purchase price	RMB 190,000–210,000	RMB 150,000–170,000
Powertrain	Battery electric vehicle (BEV)	Gasoline vehicle (ICEV)
Monthly operating cost	About RMB 600	About RMB 900
Daily energy replenishment convenience	moderate	Easy
Smart cockpit features	Advanced	Advanced
Long-term reliability	High	High
Cabin comfort and entertainment experience	Basic	Basic

Which vehicle would you choose?

- ☐ A
- ☐ B

TA4 (T04)

Attribute	Vehicle A	Vehicle B
Purchase price	RMB 150,000–170,000	RMB 190,000–210,000
Powertrain	Battery electric vehicle (BEV)	Gasoline vehicle (ICEV)
Monthly operating cost	About RMB 300 per month	About RMB 900 per month

Daily energy replenishment convenience	moderate	Easy
Smart cockpit features	Intermediate	Intermediate
Long-term reliability	High	High
Cabin comfort and entertainment experience	Premium	Premium

Which vehicle would you choose?

- ☐ A
- ☐ B

TA5 (T05)

Attribute	Vehicle A	Vehicle B
Purchase price	RMB 190,000–210,000	RMB 190,000–210,000
Powertrain	Battery electric vehicle (BEV)	Gasoline vehicle (ICEV)
Monthly operating cost	About RMB 300 per month	About RMB 300 per month
Daily energy replenishment convenience	moderate	moderate
Smart cockpit features	Intermediate	Basic
Long-term reliability	Low	High
Cabin comfort and entertainment experience	Improved	Premium

Which vehicle would you choose?

- ☐ A
- ☐ B

TA6 (T06)

Attribute	Vehicle A	Vehicle B
Purchase price	RMB 110,000–130,000	RMB 150,000–170,000
Powertrain	Battery electric vehicle (BEV)	Gasoline vehicle (ICEV)
Monthly operating cost	About RMB 300 per month	About RMB 900 per month
Daily energy replenishment convenience	easy	Easy
Smart cockpit features	Basic	Intermediate
Long-term reliability	Medium	Medium
Cabin comfort and entertainment experience	Premium	Premium

Which vehicle would you choose?

- ☐ A
- ☐ B

TA7 (T07)

Attribute	Vehicle A	Vehicle B
Purchase price	RMB 110,000–130,000	RMB 190,000–210,000
Powertrain	Battery electric vehicle (BEV)	Gasoline vehicle (ICEV)
Monthly operating cost	About RMB 600 per month	About RMB 600 per month
Daily energy replenishment convenience	moderate	easy
Smart cockpit features	Intermediate	Advanced
Long-term reliability	Low	Low
Cabin comfort and entertainment experience	Premium	Premium

Which vehicle would you choose?

- ☐ A
- ☐ B

TA8 (T08)

Attribute	Vehicle A	Vehicle B
Purchase price	RMB 190,000–210,000	RMB 110,000–130,000
Powertrain	Battery electric vehicle (BEV)	Gasoline vehicle (ICEV)
Monthly operating cost	About RMB 600 per month	About RMB 900 per month
Daily energy replenishment convenience	Difficult	Easy
Smart cockpit features	Basic	Basic
Long-term reliability	High	High
Cabin comfort and entertainment experience	Improved	Improved

Which vehicle would you choose?

- ☐ A
- ☐ B

TA9 (T09)

Attribute	Vehicle A	Vehicle B
Purchase price	RMB 150,000–170,000	RMB 150,000–170,000
Powertrain	Battery electric vehicle (BEV)	Gasoline vehicle (ICEV)

Monthly operating cost	About RMB 600 per month	About RMB 900 per month
Daily energy replenishment convenience	Easy	Moderate
Smart cockpit features	Basic	Basic
Long-term reliability	High	Low
Cabin comfort and entertainment experience	Basic	Premium

Which vehicle would you choose?

- ☐ A
- ☐ B

TA10 (T10)

Attribute	Vehicle A	Vehicle B
Purchase price	RMB 190,000–210,000	RMB 110,000–130,000
Powertrain	Battery electric vehicle (BEV)	Gasoline vehicle (ICEV)
Monthly operating cost	About RMB 600 per month	About RMB 900 per month
Daily energy replenishment convenience	Moderate	Easy
Smart cockpit features	Basic	Intermediate
Long-term reliability	Medium	Low
Cabin comfort and entertainment experience	Improved	Improved

Which vehicle would you choose?

- ☐ A
- ☐ B

Block B

TB1 (T01)

Attribute	Vehicle A	Vehicle B
Purchase price	RMB 150,000–170,000	RMB 150,000–170,000
Powertrain	Battery electric vehicle (BEV)	Gasoline vehicle (ICEV)
Monthly operating cost	About RMB 300 per month	About RMB 600 per month
Daily energy replenishment convenience	Difficult	Moderate
Smart cockpit features	Intermediate	Intermediate
Long-term reliability	High	High
Cabin comfort and entertainment experience	Premium	Premium

Which vehicle would you choose?

- ☐ A
- ☐ B

TB2 (T02)

Attribute	Vehicle A	Vehicle B
Purchase price	RMB 190,000–210,000	RMB 150,000–170,000
Powertrain	Battery electric vehicle (BEV)	Gasoline vehicle (ICEV)
Monthly operating cost	About RMB 300 per month	About RMB 600 per month
Daily energy replenishment convenience	Difficult	Easy
Smart cockpit features	Advanced	Intermediate
Long-term reliability	Medium	Medium
Cabin comfort and entertainment experience	Basic	Basic

Which vehicle would you choose?

- ☐ A
- ☐ B

TB3 (T03)

Attribute	Vehicle A	Vehicle B
Purchase price	RMB 190,000–210,000	RMB 150,000–170,000
Powertrain	Battery electric vehicle (BEV)	Gasoline vehicle (ICEV)
Monthly operating cost	About RMB 300 per month	About RMB 600 per month
Daily energy replenishment convenience	Easy	Moderate
Smart cockpit features	Intermediate	Basic
Long-term reliability	High	High
Cabin comfort and entertainment experience	Improved	Improved

Which vehicle would you choose?

- ☐ A
- ☐ B

TB4 (T04)

Attribute	Vehicle A	Vehicle B
Purchase price	RMB 150,000–170,000	RMB 190,000–210,000
Powertrain	Battery electric vehicle (BEV)	Gasoline vehicle (ICEV)
Monthly operating cost	About RMB 600 per month	About RMB 300 per month
Daily energy replenishment convenience	Easy	Moderate

Smart cockpit features	Advanced	Advanced
Long-term reliability	Low	Medium
Cabin comfort and entertainment experience	Improved	Improved

Which vehicle would you choose?

- ☐ A
- ☐ B

TB5 (T05)

Attribute	Vehicle A	Vehicle B
Purchase price	RMB 150,000–170,000	RMB 110,000–130,000
Powertrain	Battery electric vehicle (BEV)	Gasoline vehicle (ICEV)
Monthly operating cost	About RMB 300 per month	About RMB 900 per month
Daily energy replenishment convenience	Moderate	Moderate
Smart cockpit features	Intermediate	Basic
Long-term reliability	Medium	Medium
Cabin comfort and entertainment experience	Basic	Basic

Which vehicle would you choose?

- ☐ A
- ☐ B

TB6 (T06)

Attribute	Vehicle A	Vehicle B
Purchase price	RMB 110,000–130,000	RMB 190,000–210,000
Powertrain	Battery electric vehicle (BEV)	Gasoline vehicle (ICEV)
Monthly operating cost	About RMB 600 per month	About RMB 600 per month
Daily energy replenishment convenience	Difficult	Moderate
Smart cockpit features	Intermediate	Intermediate
Long-term reliability	Low	Low
Cabin comfort and entertainment experience	Basic	Basic

Which vehicle would you choose?

- ☐ A
- ☐ B

TB7 (T07)

Attribute	Vehicle A	Vehicle B
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Purchase price	RMB 110,000–130,000	RMB 110,000–130,000
Powertrain	Battery electric vehicle (BEV)	Gasoline vehicle (ICEV)
Monthly operating cost	About RMB 600 per month	About RMB 300 per month
Daily energy replenishment convenience	Easy	Moderate
Smart cockpit features	Advanced	Intermediate
Long-term reliability	Medium	High
Cabin comfort and entertainment experience	Premium	Basic

Which vehicle would you choose?

- ☐ A
- ☐ B

TB8 (T08)

Attribute	Vehicle A	Vehicle B
Purchase price	RMB 190,000–210,000	RMB 110,000–130,000
Powertrain	Battery electric vehicle (BEV)	Gasoline vehicle (ICEV)
Monthly operating cost	About RMB 300 per month	About RMB 300 per month
Daily energy replenishment convenience	Easy	Easy
Smart cockpit features	Intermediate	Basic
Long-term reliability	Low	Low
Cabin comfort and entertainment experience	Basic	Basic

Which vehicle would you choose?

- ☐ A
- ☐ B

TB9 (T09)

Attribute	Vehicle A	Vehicle B
Purchase price	RMB 150,000–170,000	RMB 190,000–210,000
Powertrain	Battery electric vehicle (BEV)	Gasoline vehicle (ICEV)
Monthly operating cost	About RMB 300 per month	About RMB 900 per month

Daily energy replenishment convenience	Moderate	Moderate
Smart cockpit features	Basic	Advanced
Long-term reliability	Low	Medium
Cabin comfort and entertainment experience	Basic	Basic

Which vehicle would you choose?

- ☐ A
- ☐ B

TB10 (T10)

Attribute	Vehicle A	Vehicle B
Purchase price	RMB 110,000–130,000	RMB 190,000–210,000
Powertrain	Battery electric vehicle (BEV)	Gasoline vehicle (ICEV)
Monthly operating cost	About RMB 300 per month	About RMB 600 per month
Daily energy replenishment convenience	Difficult	Easy
Smart cockpit features	Basic	Basic
Long-term reliability	Low	Medium
Cabin comfort and entertainment experience	Premium	Premium

Which vehicle would you choose?

- ☐ A
- ☐ B

4. Demographic Information

Q3. Age:

- ☐ 18–24
- ☐ 25–29
- ☐ 30–39
- ☐ 40 and above

Q4. Gender:

- ☐ Male
- ☐ Female
- ☐ Prefer not to say

Q5. Education level:

- ☐ Junior college or below
- ☐ Bachelor's degree
- ☐ Master's degree or above

Q6. Monthly personal income (before tax):

- ☐ Below 8,000 RMB
- ☐ 8,001–15,000 RMB
- ☐ 15,001–25,000 RMB
- ☐ Above 25,000 RMB

Q7. City tier of residence:

- ☐ Tier 1 / New Tier 1 cities
- ☐ Tier 2 cities
- ☐ Tier 3 and below
- ☐ County / rural areas

Q8. Do you or your household own a car?

- ☐ Yes
- ☐ No

Q9. Do you have access to home charging installation?

- ☐ Yes
- ☐ No

Q10. One-way daily commuting distance:

- ☐ No regular commute
- ☐ Within 10 km
- ☐ 11–30 km
- ☐ Above 30 km

Q11. Do you have experience using electric vehicles (EV)?

- ☐ Yes
- ☐ No